

CS781 — Proposed Papers for Presentation

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Why I chose these papers

For class, I want to present **two “anchor” papers**:

- **One classic, highly cited EHR paper** that established a practical baseline for representation learning on longitudinal clinical records.
- **One newer genomics paper** that shows what modern deep learning looks like when the “input” is biological sequence and the goal is functional prediction.

That combo gives me two major modalities the course emphasizes (**clinical records + biological sequence**) and it also lines up with my current work in the lab on **All of Us (AoU) respiratory disease analysis**.

Note: Everything listed here is **primary research** (original empirical studies with new methods/results), not review articles.

Papers I’m ready to present (my top two)

1) Miotto et al. (2016) — *Deep Patient*

Area: EHR representation learning / clinical prediction

Why I picked it: This is one of the early papers that made “learn an embedding from longitudinal EHR codes and use it for prediction” feel concrete and useful. It’s also a great paper for talking about the stuff that can quietly break clinical ML: coding drift, site bias, temporal leakage, and confounding.

Why it fits my work: I can translate the ideas directly to AoU-style cohort work (stratifying patients and predicting/characterizing outcomes related to respiratory disease).

How I’ll present it (what I’ll focus on):

- How the unsupervised patient representations are learned from longitudinal EHR data
- Evaluation setup and the common failure modes (leakage, confounding, site effects)
- How I’d adapt the pattern to AoU in a responsible way (cohort definition, validation strategy, and what “good” looks like)

2) Avsec et al. (2021) — *Enformer*

Area: Regulatory genomics / sequence → function

Why I picked it: I want one talk that’s *not* EHR, and Enformer is a strong modern example of deep learning applied to genomics where the key challenge is **long-range context** in DNA. The paper is also very teachable because it’s clear about inputs/outputs, benchmarks, and the limits of interpretation.

Why it fits my work: If I extend respiratory work beyond phenotype/EHR into genotype/regulatory interpretation (or even just want a serious reference point for sequence modeling in biomed), this paper is a strong anchor.

How I'll present it (what I'll focus on):

- Why long-range context matters for gene regulation
 - What the model predicts, what the training data looks like, and how performance is measured
 - Variant-effect prediction: what you can infer vs what you *can't* infer, especially for clinical translation
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Other papers I can present if needed (backup options)

3) Jumper et al. (2021) — *AlphaFold*

Area: Protein structure prediction

Why I'd present it: It changed the field and it's a clean way to discuss evaluation (CASP), generalization, and what "breakthrough performance" means in practice.

What I'd focus on: The core idea of the architecture at a high level, how it was evaluated, and the difference between scientific utility and clinical validation.

4) Ronneberger et al. (2015) — *U-Net*

Area: Biomedical image segmentation

Why I'd present it: U-Net is the foundation for a lot of medical imaging segmentation work and it's very teachable.

What I'd focus on: Encoder-decoder + skip connections, segmentation losses, and the real-world issues (label quality and domain shift).

5) Esteva et al. (2017) — Skin cancer classification

Area: Clinical imaging classification

Why I'd present it: Strong example for discussing study design, dataset shift, calibration, and what it takes to make "clinician-level" claims meaningful.

What I'd focus on: How the comparisons are set up and what you'd need to reproduce/translate it into a real workflow.

6) Lee et al. (2020) — *BioBERT*

Area: Biomedical NLP

Why I'd present it: It's a key bridge paper: it shows domain-adaptive pretraining and sets up the path toward modern clinical NLP and today's LLM ecosystem.

What I'd focus on: What domain pretraining buys you, how biomedical NLP tasks are evaluated, and what changes when you move from BioBERT-era models to current LLMs.

Optional: I can also present my own project direction

If it's useful for the class, I can present a short talk on my lab-aligned direction:

- Exploratory respiratory disease analysis in AoU (COPD/asthma/allergies + SDOH)
 - A forward-looking privacy-preserving / decentralized AI framing (TinyLLM-style ideas) as a roadmap
 - Potential integration of environmental factors via Google Earth Engine / AlphaEarth-style embeddings as a methodological extension (without applying an LLM directly to participant-level AoU data at this stage)
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References (APA)

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